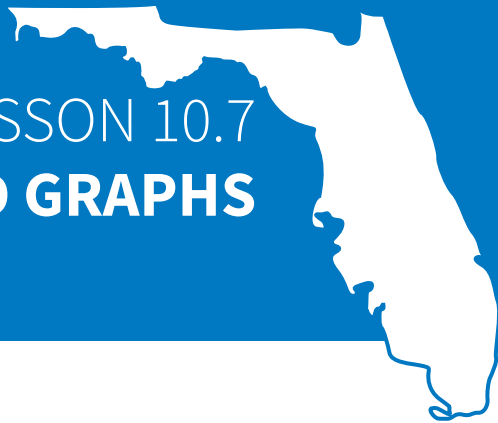


LESSON 10.7

ANALYZING DESCRIPTIONS AND GRAPHS



LESSON INTRODUCTION

MA.8.F.1.3 Analyze a real-world written description or graphical representation of a functional relationship between two quantities and identify where the function is increasing, decreasing or constant.

LEARNING TARGETS

- I can analyze a graph representing a real-world functional relationship.
- I can analyze a written description of a real-world functional relationship.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How do you write a description of the functional relationship between the independent and dependent variables seen in a graph?
- What strategies do you use to analyze and sketch graphs of functional relationships?
- How do you determine where a function is increasing, decreasing, or constant?

LESSON NARRATIVE

This lesson continues to analyze real-world written descriptions along with their matching graphical representations. Using clues from both the graph and/or the written descriptions, students will engage in a Card Sort as well as write descriptions for given graphs. Students summarize and discuss helpful tips and strategies for analyzing and sketching graphs of functional relationships with their classmates.

COMPONENT SUMMARY

10.7.1 WARM-UP (~5 MIN.)

Determine which of three stories matches a graph

10.7.2 ACTIVITY (10-15 MIN.)

Card Sort matching graphs to stories

10.7.3 BRING IT TOGETHER (10 MIN.)

Summarize and discuss tips and strategies for analyzing and sketching graphs of functional relationships

10.7.4 YOUR TURN! (10-15 MIN.)

Analyze graphs to write stories and identify features

10.7.5 WRAP-UP (~5 MIN.)

Reflect on understanding from the lesson

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- Card Sort
- (Optional) Poster paper
- (Optional) Markers
- (Optional) Glue

ADDITIONAL PREPARATION

- Print the stories and graph cards out for students to match for 10.7.2 Activity. You can either provide students with scissors or cut the cards out in advance and give them to the students in envelopes or Ziploc bags.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may interpret flat lines as directional rather than the relationship between height and time or distance and time.
- Students may think the dependent variable of a graph is speed when it is distance.

THINGS TO CONSIDER

- Create an anchor chart and allow students to create reference cards with definitions, instructions, and examples to aid them in analyzing the graphs of functional relationships.
- When analyzing graphs, consider writing explanations next to appropriate sections and key features on the graph to make visual connections to the stories.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

Frame the interpretations of graphs using a Notice and Wonder to provide a low-floor, high-ceiling that encourages all responses as correct and simultaneously extends students' thinking beyond what might be correct or incorrect in a given context.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Affirm, recognize, and praise student work and response language that reflects strong math thinking and reasoning in MA.K12.MTR.7.1 to facilitate a positive learning environment.

- “Question the accuracy of models and methods...”
- “Validate conclusions by comparing to a given situation...”
- “Redesign models and methods to improve accuracy...”

DIFFERENTIATE INSTRUCTION

Consider promoting students to write observations to highlight key features on graphs as additional scaffolding for interpreting the story of each graph. Students highlighting features like y-intercept might inform more precise stories. Notice the intentional differentiation throughout the lesson. The variety in processing provides multiple entry points for activating learning using visual, verbal, or kinesthetic methods. Make space for each to engage all learners throughout the lesson. Utilize 10.7.3 Bring It Together as an opportunity for students to demonstrate their learning. Responses from this activity or informal observations from the Card Sort will better inform responses in 10.7.5 Wrap-Up.

10.7.1 WARM-UP: BELL WORK

TEACHER MOVES

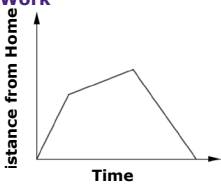
Story B presents a common misconception that associates the way the graph looks with what it is representing. The graph's shape looks like a hill that starts out steep and then flattens, but the relationship is modeling the distance from home in relation to time.



10.7.1 Warm-Up: Bell Work

Three stories are given.

One of the stories matches the graph shown.



Story A	Story B	Story C
Yadiel rode his bike toward his friend's house. After riding for a few minutes, he started riding faster until he remembered he forgot his phone, so he turned around and rode home to get it.	Yadiel and his brother rode their bikes up the hill on their street. The hill is steep before it flattens out. Once they got to the top, they rode smoothly down the other side to their neighbor's house.	Yadiel rode his bike quickly to meet up with his neighbor. Once they met up, they rode a little bit slower to meet up with another friend before turning around and riding back to Yadiel's house to hang out.

- Determine which story matches the graph.
Story C matches the graph.
- Describe which clues from the graph or the story you used to match the story and the graph.
Story A: What's correct about it? He started riding away from home, his pace changed, and then he turned back around to go back home. What's incorrect about it? When his pace changed, he slowed down before turning around...he didn't speed up. Story B: What's correct about it? He could have been riding up a hill and something changed after a little bit... What's incorrect about it? This story is describing the height they are up the hill, but the graph is meant to be about distance from home. Also, he went back home by the end of the graph, not to a different location because the y-axis shows the distance from his home.

 10.7.2 Activity: Card Sort

Work with your partner on the following.

1. Match the cards. Write the number of the interpretation that matches the graph.

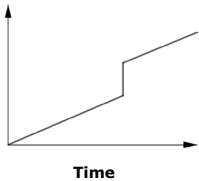
Graph	Interpretation
A	6
B	3
C	4
D	2
E	1
F	5

2. For the graph that matches Interpretation Card 6, write a story that matches this graph.
Answers will vary. Responses should include someone traveling away from home at a constant pace, turning around to go back at a slower pace for some reason, and then turning back around to travel away from home again at the same pace as the beginning.

3. A graph that shows the distance from home versus time is shown.

Alexandra says the relationship represented by the graph is not possible. Explain why.

The relationship is not possible because at a certain point in time the graph is increasing distance from home and not changing in time. It is impossible for time to stand still while the distance is changing.



10.7.2 ACTIVITY: CARD SORT

DIFFERENTIATE INSTRUCTION

Facilitate the Card Sort using informal observations and decisions that best fit the needs of your classroom.

Partner Roles: Students can work in collaboration, or you can offer constraints to differentiate to engage all learners. For example, one partner can only talk, while the other partner is the only one who can move cards. Specific roles activate different entry points and foster increased precision in mathematical discourse.

Gallery Walk: Students match cards on chart paper and write at least one sentence of rationale. Students then visit other charts and add more details where necessary.

TEACHER MOVES

Notice the opportunity for students to make sense of information and learning from the Card Sort in 10.7.3 Bring it Together. Provide space for mistakes and student-centered learning in 10.7.2 Activity Card Sort.

10.7.3 BRING IT TOGETHER: ANALYZING GRAPHS REPRESENTING REAL-WORLD CONTEXTS

ENRICHMENT

- What strategies or tips have you used to analyze functional relationships in the past?
- What strategies or tips have helped you sketch graphs of functional relationships in the past?
- Pretend you have a friend that is struggling to analyze and sketch functional relationships. What advice would you give them?



10.7.3 Bring It Together: Analyzing Graphs Representing Real-World Contexts

Consider some of the graphs and stories you have analyzed over the last few lessons.

1. What are some strategies you have learned and key tips that are helpful for you to keep in mind when analyzing or sketching graphs of functional relationships?

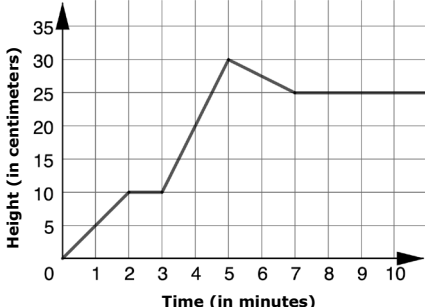
Helpful Tips	Strategies
<i>Answers will vary. Possible answers include:</i> • <i>In a distance-time graph, a horizontal line is no change in distance.</i> • <i>The independent variable of time cannot occur in two places at one time.</i>	<i>Answers will vary. Possible answers include:</i> • <i>Underline time</i> • <i>Highlighting changes in rate of change</i> • <i>Circling key parts of the context such as away and towards.</i>

2. Compare and discuss your list with your partner. Add any tips or strategies your partner shares that you find helpful to your list.



10.7.4 Your Turn!

1. Emma fills her aquarium with water. The graph shows the height of the water, in cm, in the aquarium as a function of time, in minutes.



Write a story to match the graph that describes how Emma fills the aquarium.

Answers will vary. Emma started to fill the aquarium at a constant rate for 2 minutes. Then for 1 minute there was no water flowing into the tank. For the next 2 minutes, it quickly filled to 30 cm. The tank started draining at a constant rate and after 2 minutes it remained at a constant height.

10.7.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Allow students to create and use reference cards with definitions, instructions, and examples to aid them in analyzing the graphs of functional relationships.

Consider allowing students to share their answers verbally by discussing them with a partner instead of or before providing a written explanation.

Help students understand that height is similar to the distance away from a point. Fill up a clear plastic cup steadily, then stop, then fill it up faster. Then stop and ask students how to change the water so that the height of the water in the cup is smaller.

ENRICHMENT

Questions to help students understand the relationship between height and time:

- What is happening to the height of the water between 2 and 3 minutes?
- What is the rate of change between 0 and 2 minutes and between 3 and 5 minutes?
- What is happening to the height of the water between 5 and 6 minutes?
- How do speed and velocity apply to the graph?

10.7.4 YOUR TURN! (CONTINUED)

DIFFERENTIATE INSTRUCTION

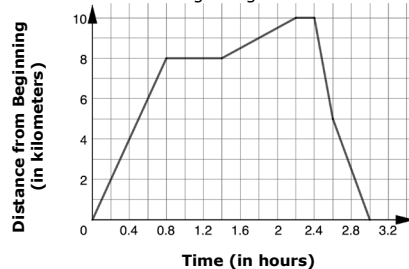
Write explanations next to appropriate sections and key features on the graph to make a visual connection to the story the graph is telling.

Having students trace the path of the graph with their finger may help them analyze the functional relationships.

DIFFERENTIATE INSTRUCTION

Encourage students to look back through their books or notes for vocabulary that was previously introduced that they may have forgotten.

2. Mason was riding his bike along the Lake Okeechobee Scenic Trail. The graph shows his trip. The trail has markers every 0.5 kilometers (km), showing the distance from the beginning of the trail.

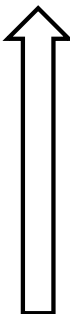


- What is the domain of Mason's bike ride?
The domain of Mason's bike ride is $0 \leq x \leq 3$.
- What is the range of Mason's bike ride?
The range of Mason's bike ride is $0 \leq y \leq 10$.
- Between which times was Mason going the fastest?
Mason was going the fastest between 2.4 and 2.6 hours with a rate of 25 km/h.
- Between which times was Mason riding away from the beginning of the trail?
Mason was riding away from the beginning of the trail between 0 to 0.8 hours and 1.4 to 2.2 hours.
- Between which times was Mason riding back to the beginning of the trail?
Mason was riding back towards the beginning of the trail between 2.4 and 3 hours.
- Between which times was Mason stopped?
Mason was stopped between 0.8 to 1.4 hours and 2.2 to 2.4 hours.

**10.7.5 Wrap-Up: Reflection**

1. Color in the arrow up to the statement that best describes your current understanding of the benchmark below.

Analyze a real-world written description or graphical representation of a functional relationship between two quantities and identify where the function is increasing, decreasing or constant.



I'm so confident, I could explain how to analyze or sketch graphs of functional relationships to someone else!

I can get to the right answer, but I don't understand well enough to explain how to analyze or sketch graphs yet.

I understand some of this, but I don't understand all of it yet.

I have been trying hard, but I am finding this challenging.

Answers will vary.

10.7.5 WRAP-UP: REFLECTION**QUESTIONING**

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.